Assessment Assumptions, Baseline & Sensitivity Analysis

Practical guidance for preparing and testing a credible 12-month SME projection.

This guide explains how to prepare the assessment baseline and use Sensitivity Analysis to test how changes in key assumptions could affect profitability, liquidity and viability. It supports government-programme assessment and management discussion; it is not a substitute for audited accounts, professional tax advice or a guarantee of future performance.

> **Recommended sequence:** capture verified opening facts, enter reasonable baseline assumptions, review readiness, confirm the baseline, and then use Sensitivity Analysis to test uncertainty without changing the approved assessment records.

1. Opening Position

Setting	How it supports projections
Opening stock, debtors and creditors	Read-only totals calculated from active Sales Categories. Update the underlying category records rather than entering duplicate totals.
Opening bank balance	Sets the initial cash available. Every projected receipt and payment builds from this amount.
Overdraft interest	Provides the annual financing rate applied where projected cash becomes overdrawn.
Gross assets, accumulated and projected depreciation	Read-only totals calculated from active asset records and the asset depreciation engine.
Other current assets and liabilities	Captures prepayments, non-trade receivables, accruals and short-term obligations not represented by trade debtors or creditors. These remain on the projected balance sheet unless reclassified outside this lightweight model.
Opening VAT payable	VAT already owed at the start date and treated as a Month 1 cash outflow.
Opening VAT refundable	A confirmed VAT refund receivable at the start date and treated as a Month 1 cash inflow.

Evidence: recent bank statement, stock count, management accounts, aged debtors/creditors, asset register and loan statements. A genuine zero should be confirmed rather than left unknown.

2. Sales Categories

Setting	Projection use

Category and income type	Separates true Sales from sundry income, grants, disposals and other income. Only true Sales should drive gross profit, stock, debtors and break-even measures.
Markup	Supports the category cost-of-Sales estimate and gross-margin calculation when purchase/stock information is incomplete.
Opening stock	Assigns starting stock to the relevant Sales Category and feeds opening working capital.

3. Debtors & Creditors Profiles

Percentage mode Enter the expected distribution across ageing buckets. The 0–30 day bucket balances the total to 100%.

Actual values mode Enter known balances such as R90,000 in 0–30 days and R35,000 in 31–60 days. The application converts each bucket to a percentage of the total and saves those percentages for projections.

Setting	Projection use
Debtor profile	Determines when credit Sales are expected to be collected, shifting cash receipts into later months.
Creditor profile	Determines when supplier purchases are paid, shifting cash payments into later months.
Bad debt	Reduces the collectible portion of credit Sales and highlights collection risk.
Average payment days	Provides a practical reasonableness indicator for the captured collection profile.
Include VAT	Clarifies whether the entered ageing balances include VAT so cash and VAT interpretation remain consistent.

4. Sales & Stock Assumptions

Setting	Projection use
Sales growth and start month	Defines the baseline percentage change and the first month affected. “Month 4” means Months 1–3 remain at the entered baseline.
Cash Sales percentage	Identifies Sales collected in the month of sale; the balance follows the debtor profile.
Bad debts	Provides a baseline loss allowance for uncollectible credit Sales. Reconcile it with the profile setting.
Fallback cost of Sales	Used only when category markup or detailed stock/purchase data cannot provide a reliable cost basis.
Minimum closing stock	Defines a simple stock buffer and supports purchase requirements and liquidity review.

5. Expenses, VAT & Director Wages

Setting	Projection use
VAT rate	Applies to taxable Sales, purchases and expenses. Confirm registration status and the legally applicable rate.

VAT payment frequency	Controls whether net VAT cash settlement is expected monthly or every two months.
Expense increase and start month	Escalates operating expenses from the selected month without changing earlier baseline months.
Contingency	Adds a modest allowance for foreseeable but unallocated operating costs. It should not conceal known expenses.
Director wages	Number of directors multiplied by monthly wage becomes the recurring management remuneration cash outflow.
Director wage increase	Applies the expected change to director remuneration during the projection period.

6. Funding, Cash Reserve & Assets

Setting	Projection use
Loan balance, rate, term and instalment	Drive financing balances, interest expense and monthly repayment cash outflows. The loan start month determines when a new facility begins affecting the projection.
Minimum cash balance	Represents management's preferred liquidity reserve. Falling below it should trigger a warning, not be treated as additional spending capacity.
Asset opening cost and accumulated depreciation	Establish net book value at the start of the projection.
Acquisition month	Determines when a planned asset enters the balance sheet and begins affecting cash and depreciation.
Depreciation rate and method	Determine the periodic non-cash depreciation charge and closing asset value.
Disposals and movements	Adjust asset values, disposal proceeds and gains/losses in the relevant month.

7. Readiness, Notes & Confirmation

The readiness score checks whether essential opening facts, Sales Categories, monthly Sales, working-capital profiles, expenses, VAT, loans, asset depreciation and baseline assumptions are available. **Incomplete** means a required check failed. **Ready with warnings** means core data is available but an advisory item remains. **Ready** means the minimum automated checks pass.

Assumption notes should record the source, date, owner and reason for each material estimate—for example, "based on signed orders" or "average of the last six VAT returns."

Confirmation records that the baseline has been reviewed and is reasonable for the assessment. It does not prove that the forecast will occur.

> **Important cautions:** avoid double-counting assumptions already represented by monthly source data; use fallback cost of Sales only when required; distinguish Sales from grants or disposals; keep VAT treatment consistent; and rerun readiness after changing material source data.

8. Using Sensitivity Analysis

Sensitivity Analysis tests how the approved 12-month projection responds when important assumptions change. It does not edit Sales, expenses, assets, loans or any other source record. Every analysis begins with a fresh copy of the authoritative baseline and applies temporary adjustments in memory.

> **Purpose:** use the module to understand downside risk, test opportunities, estimate funding pressure and explain which assumptions have the greatest influence on viability. It should support professional judgement rather than replace it.

8.1 Before starting

1. Open the correct assessment and confirm that its opening position and monthly source data are current.
2. Review Projection Readiness and resolve material missing information.
3. Confirm that the baseline projection is reasonable before testing alternatives.
4. Select **Sensitivity Analysis** immediately after **Assessment Reports** in the assessment menu.

If Sales or expense information is incomplete, the page displays a **Draft baseline** warning. Analysis can still be run, but its conclusions must be treated as provisional.

8.2 Scenario Builder

The Scenario Builder combines one or more changes into a practical alternative case. Start with a preset or enter a name, description and individual adjustments.

Control	How to use it
Scenario name and description	Use a clear name and explain the event being tested, such as “Municipal contract delayed” or “Supplier increase from Month 4.”
Driver	Select the financial assumption to change.
Adjustment	Enter a positive value for an increase or delay and a negative value for a decrease or acceleration, according to the driver.
Effective month	Select Month 1–12 when the change begins. Earlier months remain unchanged.
Duration	Select **Ongoing** for the rest of the projection or clear it and enter a limited number of months.
Add adjustment	Add the change to the scenario. Several adjustments may be combined before calculation.
Run scenario	Calculates the scenario and compares its main results with the baseline.

Scenario presets

Preset	Question it helps answer
Sales Decline	What happens if demand is weaker than expected?
Cost Inflation	Can the enterprise absorb increases in inputs and overheads?

Slow Collections	How does a customer-payment delay affect cash?
Interest Rate Increase	How sensitive is profitability to higher finance costs?
Asset Acquisition	Can the planned capital expenditure be funded?
Combined Downside	What happens when lower Sales, higher costs and slower collections occur together?
Growth Opportunity	Does additional Sales justify the extra operating expenditure?

Presets are starting points only. Review and amend every adjustment so that it reflects the enterprise being assessed.

8.3 Understanding sensitivity drivers

Driver	Meaning and interpretation
Sales / revenue	Changes projected true Sales by the selected percentage. A value of −10% tests Sales that are 10% below baseline.
Cost of Sales — relative	Changes baseline cost of Sales by a percentage. A value of +10% makes the existing cost amount 10% higher.
Cost of Sales — percentage points	Changes cost of Sales as a share of revenue. If baseline cost is 60% of Sales, +5 percentage points tests 65%.
Operating expenses	Changes recurring operating expenses by the entered percentage for the selected period.
Bad debt	Adds percentage points to the uncollectible portion of Sales and reduces expected customer receipts.
Debtor collection timing	Moves projected customer receipts by whole months. A positive value delays receipts; a negative value accelerates them.
Creditor payment timing	Moves projected supplier payments by whole months. A positive value delays payment; consider whether suppliers would permit it.
Loan interest	Changes projected loan-interest expense by a relative percentage. It does not amend contractual principal repayments.
Asset purchases	Changes baseline projected asset-purchase cash outflows by the entered percentage.
Minimum cash reserve	Sets the management buffer used to identify funding shortfalls. It is a risk threshold, not an additional expense.

> **Percentage versus percentage points:** these are different. Increasing a 60% cost-of-Sales ratio by 10% produces 66%; increasing it by 10 percentage points produces 70%.

8.4 Reading Scenario Builder results

Measure	What to assess
Revenue	Total true Sales over Months 1–12 after scenario adjustments.
Gross profit	Revenue less cost of Sales. Review both the amount and gross-margin percentage.
Profit before tax (PBT)	Profit after operating expenses, depreciation and interest, before tax.
Closing cash	Cash remaining at Month 12. A positive closing balance can still conceal an earlier shortfall.
Minimum cash	The lowest monthly closing cash balance and the month in which it occurs.

Variance from baseline	The amount by which the scenario result improves or deteriorates relative to the approved baseline.

The deterministic explanation lists the largest adjustments and the resulting movement in profit and closing cash. Expand **Calculation scope** to review modelling limitations.

8.5 Sensitivity Results

This tab changes one driver at a time while holding all others at baseline. It is the best place to identify which assumption has the greatest effect.

1. Select the driver.
2. Enter the minimum and maximum adjustment.
3. Enter the step between test points. For example, −30 to +20 with a step of 5 tests −30%, −25%, −20% and so on.
4. Select the effective month and choose **Run range**.
5. Compare the baseline row with each test point and review the status colour and warning flags.

The chart shows closing cash across the range. The table adds revenue, gross margin, EBITDA, PBT, minimum cash, funding gap, current ratio and interest cover. Use **CSV** to download the detailed results for further review.

8.6 Scenario Comparison

This tab compares the baseline with up to three named scenarios. Select the scenarios to display their KPI variances and M1–M12 revenue, PBT and closing-cash movements.

- Review amount and percentage variances, not only final totals.
- Identify the first month in which cash or profit becomes unacceptable.
- Compare the cause descriptions to distinguish operational risk from financing risk.
- Export the comparison to CSV when the evidence must be shared or retained.

8.7 Risk & Break-even

Enter management's minimum cash reserve and recalculate. The module progressively tests the baseline to estimate practical limits.

Threshold	Interpretation
Maximum Sales decline	The first tested decline that produces a loss or breaches the selected cash reserve.
Maximum operating-expense increase	The first tested increase that produces a loss or breaches the reserve.
Minimum gross margin	An approximate annual margin required to cover operating expenses, depreciation and interest.
Collection-delay tolerance	The first whole-month customer-payment delay that breaches the reserve.
Minimum projected cash	The lowest monthly cash balance in the baseline and the month in which it occurs.

Findings are prioritised and paired with a practical management action. They are prompts for investigation and must be considered together with contracts, security, funding terms and management capacity.

8.8 Status colours

Status	Meaning	Suggested response
Green	No tested key threshold is breached.	Continue monitoring and retain contingency plans.
Amber	A cash reserve or coverage warning exists.	Investigate the affected month and agree a mitigation before approval.
Red	The scenario produces negative cash or a projected loss.	Do not rely on the case without corrective action, additional funding or revised assumptions.

8.9 Printing and exports

Select **Print / Save PDF** to open the browser print dialogue and save the active analysis view as a PDF. The Sensitivity Results and Scenario Comparison tabs also provide CSV exports. Record the assessment, baseline date, scenario name, adjustments and calculation date when sharing results.

8.10 Important calculation limitations

> - Scenarios are held in memory for the current page session and are not saved to the database.

- The baseline remains unchanged; update source data on the appropriate assessment page when a change is intended to become permanent.
- Existing VAT schedules remain at baseline during sensitivity overlays.
- Collection or supplier payments shifted beyond Month 12 fall outside the analysis horizon.
- Asset-purchase sensitivity changes the projected purchase cash outflow; it does not rebuild a new detailed asset register or depreciation schedule.
- Thresholds are tested estimates and may be rounded to the selected increment or whole month.
- Sensitivity results show what could happen under stated assumptions; they do not assign probabilities or predict which scenario will occur.

8.11 Recommended sensitivity review

1. Run the baseline and confirm that no material data is missing.
2. Run a Sales range to understand demand risk.
3. Run cost-of-Sales and operating-expense ranges to understand margin pressure.
4. Test slower collections and review the minimum-cash month.
5. Run the Combined Downside preset and adapt it to the enterprise.
6. Compare baseline, optimistic and pessimistic cases.
7. Set the required cash reserve and review risk thresholds.
8. Document management actions, supporting evidence and unresolved uncertainties.
9. Export or print the final analysis for the assessment record.

9. AI Analysis (Demo)

The optional fifth Sensitivity Analysis tab asks Google Gemini to interpret an already calculated baseline or scenario. The deterministic projection remains authoritative: AI does not calculate, alter or persist financial figures.

> **Privacy and free-tier disclosure:** Gemini free-tier data may be used by Google to improve its products. Only calculated financial metrics, M1–M12 aggregates, deterministic flags and numeric scenario adjustments are sent. The application excludes the business name, owner, customers, suppliers, notes, user identity, assessment ID, database entities and source records. Do not enable the demo where this processing is unacceptable.

9.1 Consent and use

1. Open the required assessment and navigate to **Sensitivity Analysis**.
2. In **Scenario Builder**, select a preset or enter custom adjustments, then select **Run scenario**.
3. Review the deterministic results first. These figures remain the authoritative financial analysis.
4. Open **AI Analysis (Demo)**. If no adjusted scenario has been run, the AI analyses the baseline. Otherwise, it analyses the current adjusted scenario against that baseline.
5. Confirm that the provider area displays **Google Gemini**, model **gemini-3.8-flash** and status **Configured**.
6. Read the Google free-tier disclosure and select the explicit consent checkbox. Consent resets when the assessment context changes.
7. Select **Generate AI Analysis**. The button remains unavailable without consent.
8. Review the executive summary and prioritised findings table, checking every recommendation against the authoritative evidence column.
9. Use **Print / Save PDF** if a point-in-time copy is required. AI results are not saved by the application.

9.2 Reading the table

Column	Meaning and control
Priority	Model-assigned review order within the configured maximum number of findings.
Area and severity	Risk theme and validated Critical, High, Medium or Low classification.
Finding	Advisory interpretation constrained to the supplied calculated data.
Authoritative Evidence	Rendered by Viability.IQ from allow-listed evidence keys. Gemini cannot supply the displayed financial values.
Recommended Action	A suggested management or assessor follow-up, not an instruction or professional opinion.
Confidence	The model's 0–100 self-rating; it is not a probability or validation score.

9.3 Recommended AI review sequence

Generate separate analyses for the cases that are material to the assessment:

1. **Baseline:** identify weaknesses already present in the expected projection.
2. **Sales Decline:** assess exposure to weaker demand.

3. **Cost Inflation:** assess pressure on gross margin and operating profit.
4. **Slow Collections:** assess working-capital and minimum-cash pressure.
5. **Combined Downside:** assess resilience when several adverse events occur together.
6. **Growth Opportunity:** assess whether additional Sales remain affordable after the associated expenditure and working-capital requirements.

Because results are not saved, print each material analysis before moving to another scenario. Running a new scenario clears the previous AI result so that commentary from one case cannot be mistaken for another.

9.4 Regenerating, clearing and interpreting results

- **Regenerate** asks Gemini to interpret the same current result again. Wording, priorities and emphasis may vary between calls.
- **Clear AI analysis** removes the displayed response from memory.
- Changing assessments clears the response and resets consent.
- Treat confidence as the model's own assessment, not as a statistical probability.
- A green or reassuring narrative does not guarantee viability.
- Investigate Critical and High findings before relying on the scenario.
- Trust the deterministic figures and authoritative evidence more than the narrative interpretation.

9.5 Server configuration

Create a key in **Google AI Studio**: sign in at `https://aistudio.google.com/app/apikey`, select or create a Google Cloud project, choose **Create API key**, then copy it to a secure secret store. Never add the key to appsettings.json or source control.

For local development from the repository root:

```
dotnet user-secrets set "SensitivityAI:ApiKey" "YOUR_KEY" --project ViabilityIQ.Web
dotnet user-secrets set "SensitivityAI:Enabled" "true" --project ViabilityIQ.Web
```

For a server, set environment variables `SensitivityAI__ApiKey` and `SensitivityAI__Enabled=true`, then restart the process. Safe defaults use endpoint `https://generativelanguage.googleapis.com/v1beta/interactions`, model `gemini-3.8-flash`, a 45-second timeout and at most six findings.

9.6 Errors and limitations

- Disabled or unconfigured states show setup guidance and never show a fabricated table.
- Credential rejection (401/403), free-tier rate limiting (429), timeout, service failure, safety blocking, cancellation and invalid structured output are reported as safe status messages.
- Unknown evidence keys, malformed JSON, invalid severity, missing or oversized text, duplicate rows and out-of-range finding counts are rejected. Raw model output is never shown.
- Free-tier quotas and rate limits are controlled by Google and can change. Availability is not guaranteed.
- This is a demonstration, not a production recommendation. Confirm every conclusion against source records, deterministic calculations and suitably qualified professional judgement.

A later provider can be introduced by implementing `ISensitivityAiAnalysisService` and changing dependency injection; the page and shared validated result contract remain provider-neutral.

Recommended overall review sequence

1. Verify the opening date and balances.
2. Check income classification and monthly Sales.
3. Confirm debtor, creditor and stock behaviour.

4. Review expenses, VAT and wages.
5. Reconcile loans and assets to supporting statements.
6. Review warnings, document sources and confirm the baseline.
7. Run sensitivity scenarios and record the principal risks and mitigations.